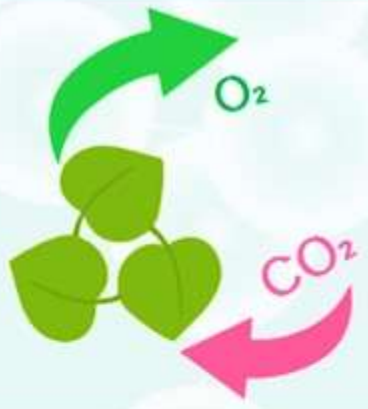


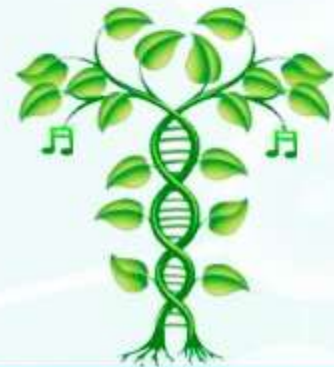


BIOLOGY



CELL

MODULE - 7



WHAT DOES A LIVING THING COMPRISES OF ?



ORGANISM



ORGAN SYSTEM



ORGANS



TISSUES



....



HOW THE SMALLEST UNIT OF LIVING THINGS GOT DISCOVERED ?



LEEUWENHOEK



ROBERT HOOKE

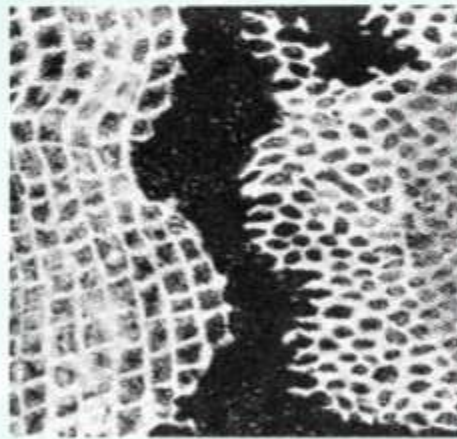
- The first microscope was constructed by Antony van Leeuwenhoek.
- His microscope consisted of a single lens and is called **simple microscope**.
- Robert Hooke, developed a microscope by using two lenses for greater magnification.
- Such microscopes was later called as **compound microscope**.



PROBING THE TINY THINGS THAT HAS LIFE !!!



Hooke's microscope

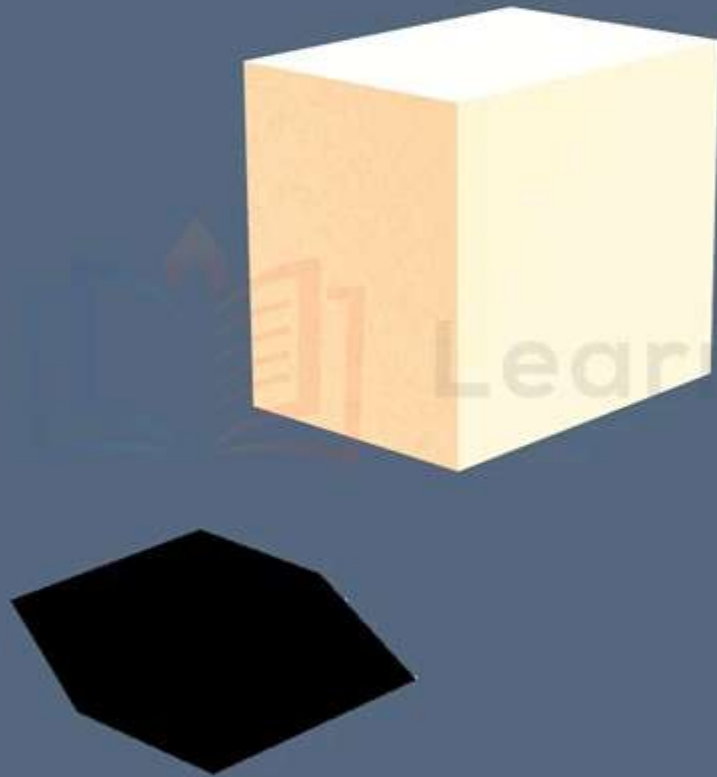


Cork cells (dead)

- Hooke examined a thin slice of cork under his microscope.
- He observed that it was made up of tiny box like compartments.
- Thus he named them cells which literally means like small rooms.
- The cells which Hooke saw were all dead and had only the empty cell walls

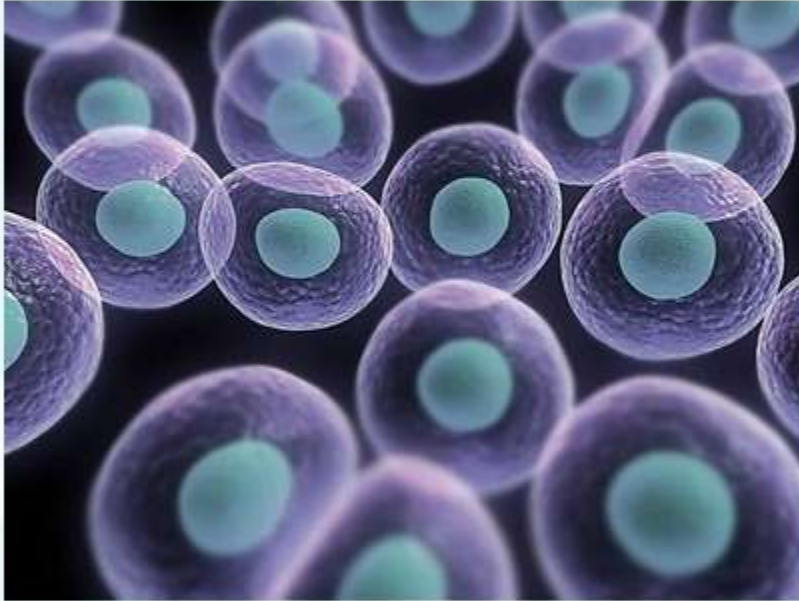


WHY A CELL NEEDS TO BE SMALL ??



WHY DO THE CELLS NEED TO BE SMALL ?

Smaller cells have larger surface area



- 1) Larger surface area will allow more nutrient to come into the cell.
- 2) More metabolic waste can be thrown out of the cell.
- 3) Respiratory gases can diffuse in and out of the cell.
- 4) Any damage to the cell can be easily repaired.



THE CELL THEORY



M. SCHLEIDEN



T. SCHWANN



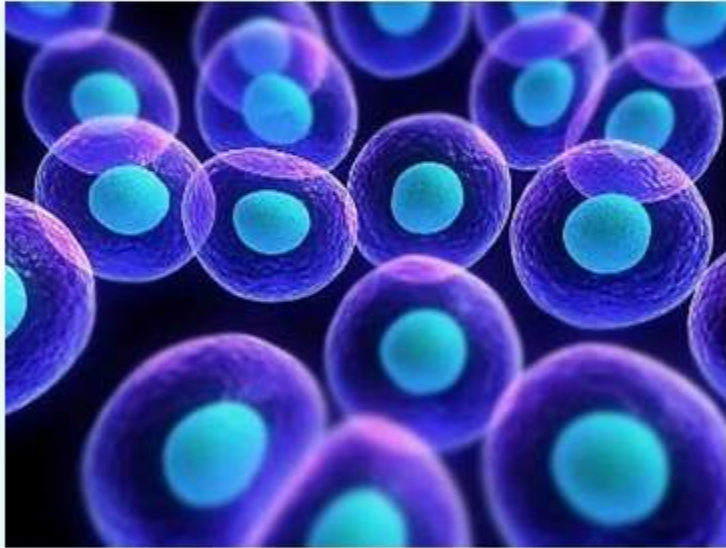
R. VIRCHOW

- Schleiden, a German botanist announced that every plant is made up of a large number of cells and these cells performed various functions.
- Schwann, a German zoologist, declared that animal and plant are made up of cells which serve as the units of structure and function.
- This was collectively called as the Cell Theory.
- Virchow made an addition to that theory by saying that all cells exist from pre-existing cells.





CELL THEORY - WHAT DOES IT MEAN?



1. Cells are the structural units of any living being.
2. Cells are the functional unit of any living being.
3. Cells die and are replaced from pre-existing cells.
4. All cells arise from one single cell !





TYPES OF MICROSCOPES

COMPOUND MICROSCOPE



The compound microscope uses glass lenses to magnify the object

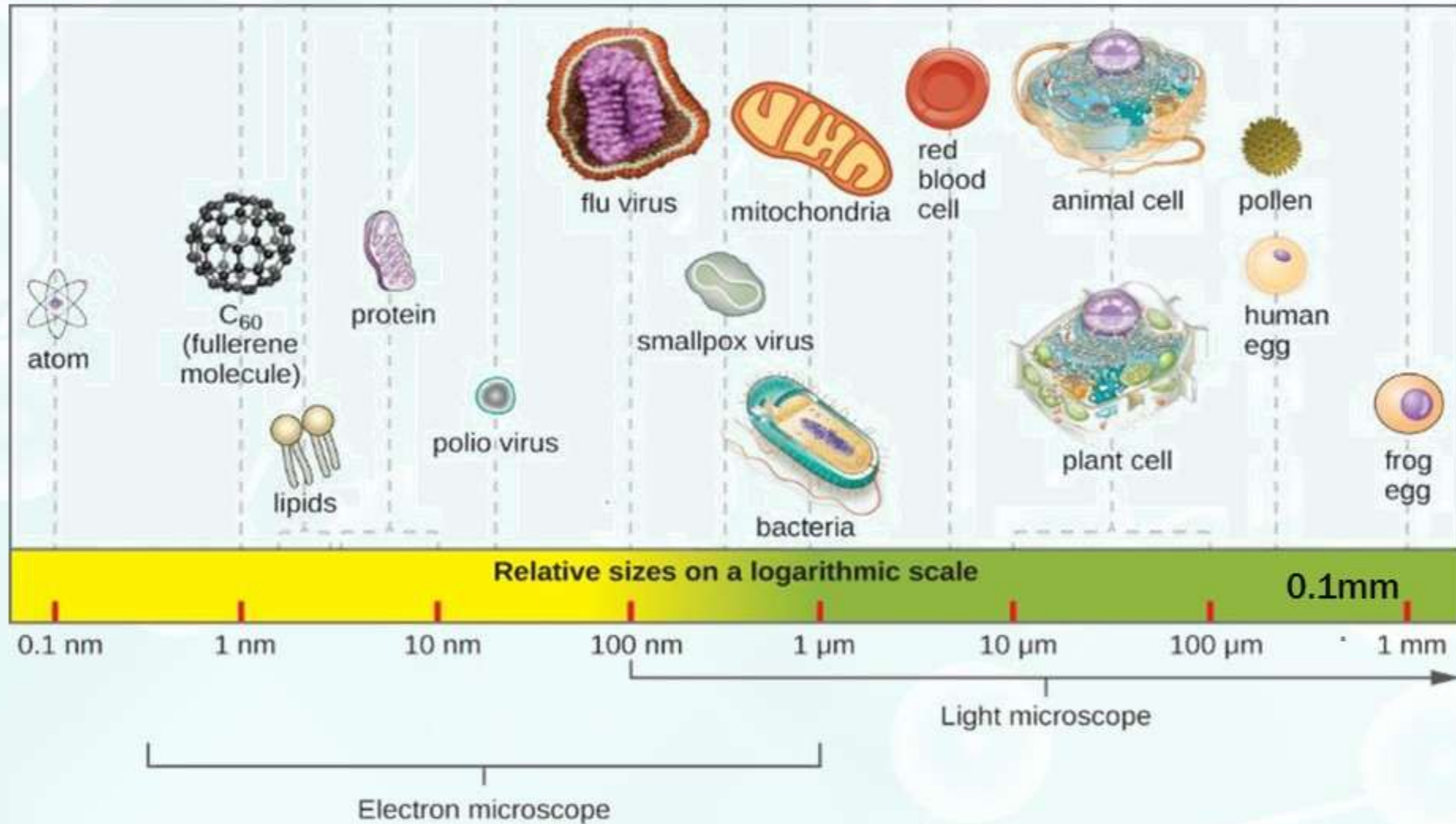
ELECTRON MICROSCOPE

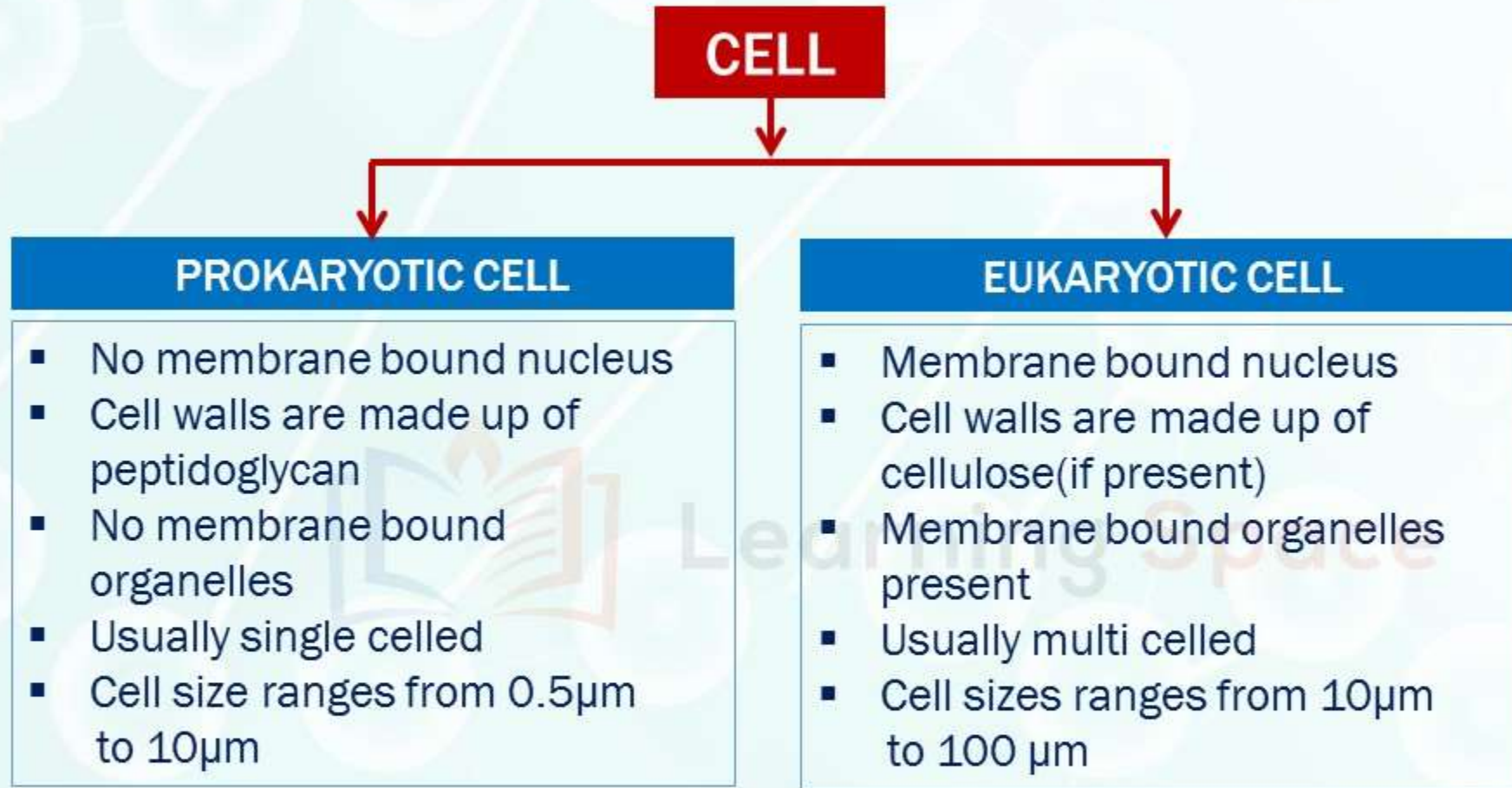


E.M uses beams of electrons which are bent by magnets.



THE SCALE OF THE CELL

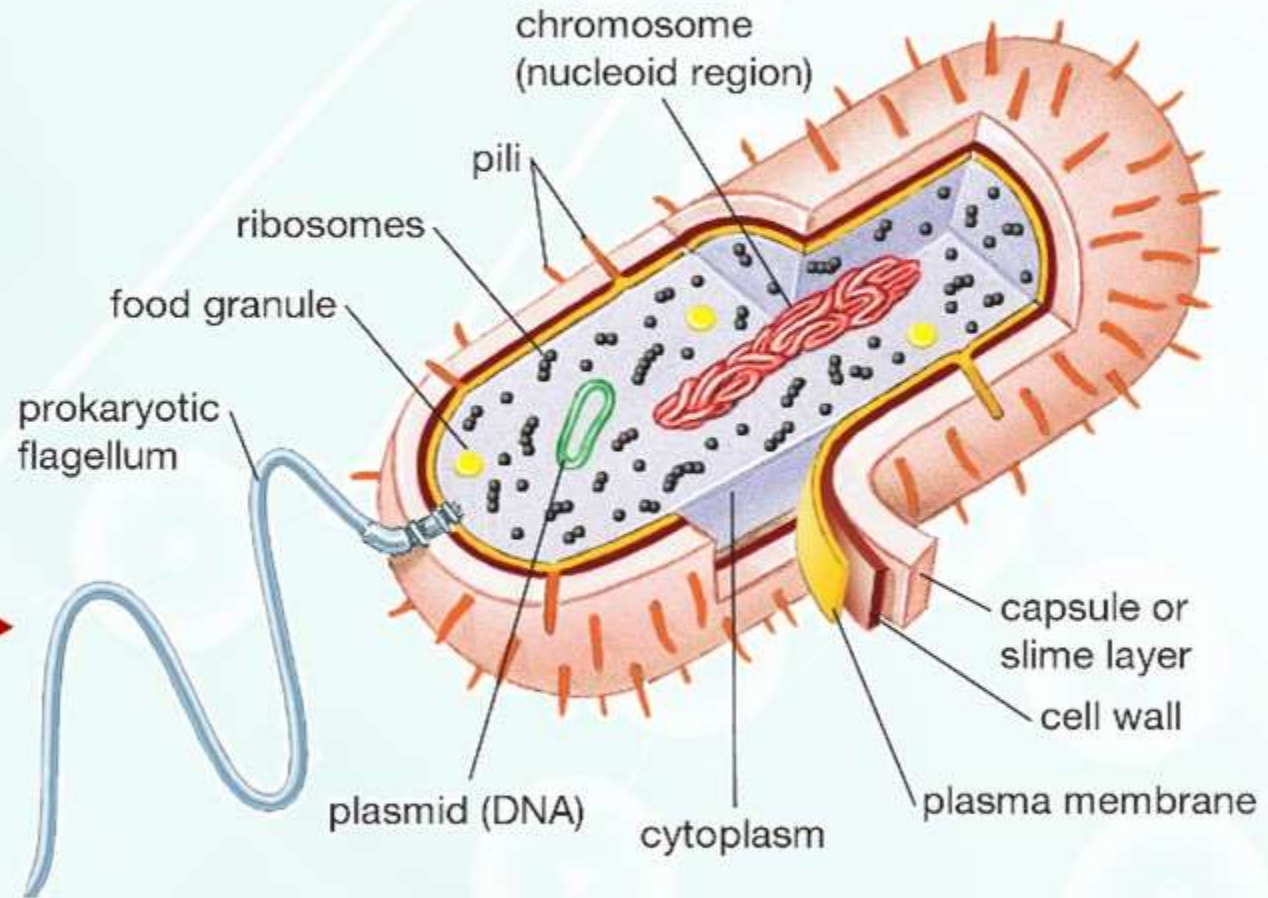
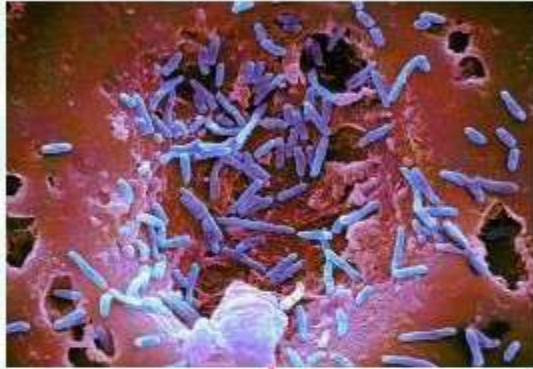




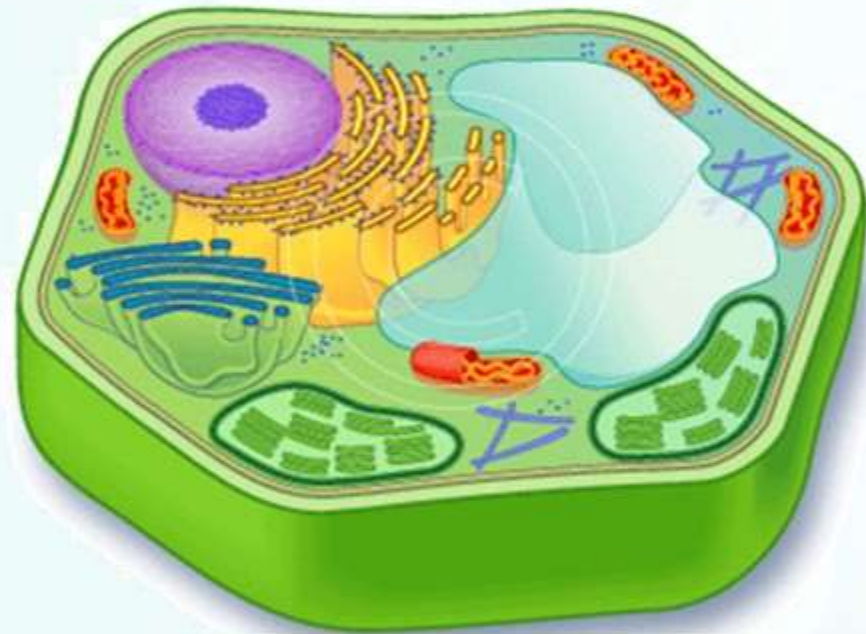
Pro- primitive, Eu- new, karyon – nucleus
The main difference between them is structural



A TYPICAL PROKARYOTIC CELL



A TYPICAL EUKARYOTIC CELL



An animal cell is usually smaller than a plant cell.



DIFFERENT SHAPES AND SIZES OF THE ANIMAL CELLS !

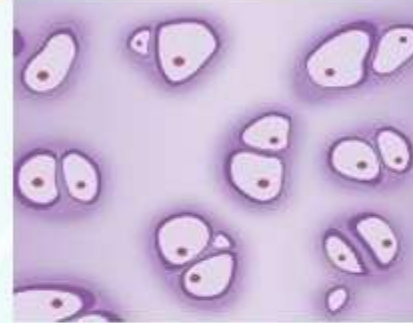
NERVE CELL



EPITHELIAL CELLS



CARTILAGE CELLS



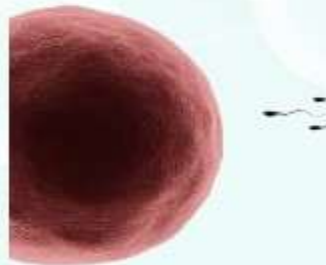
RBC



SPERM CELLS



OVUM



WBC



MUSCLE CELLS

Skeletal muscle



Smooth muscle

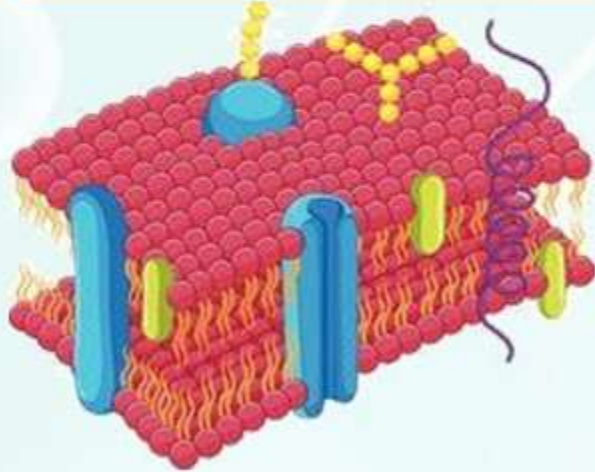


Cardiac muscle

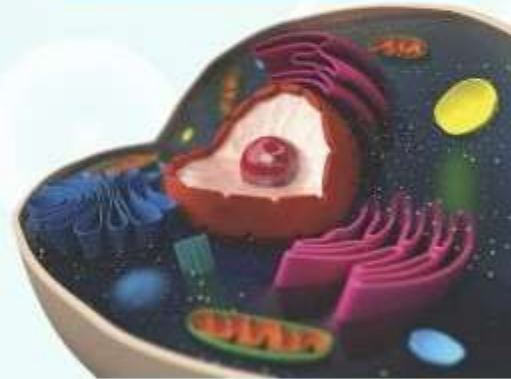


THE ULTRASTRUCTURE OF A TYPICAL CELL

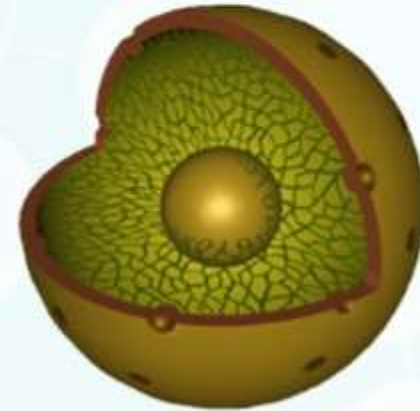
CELL MEMBRANE



CYTOPLASM

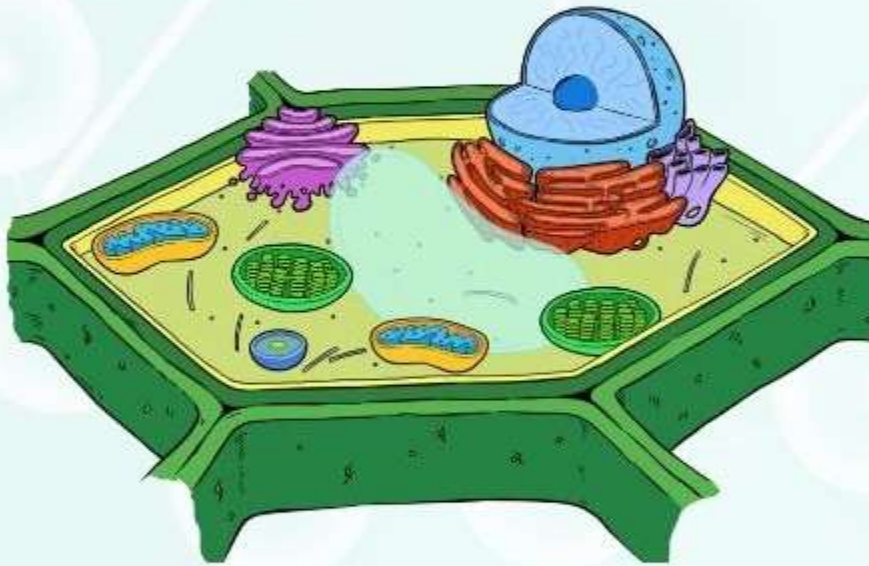


NUCLEUS





THE CELL WALL

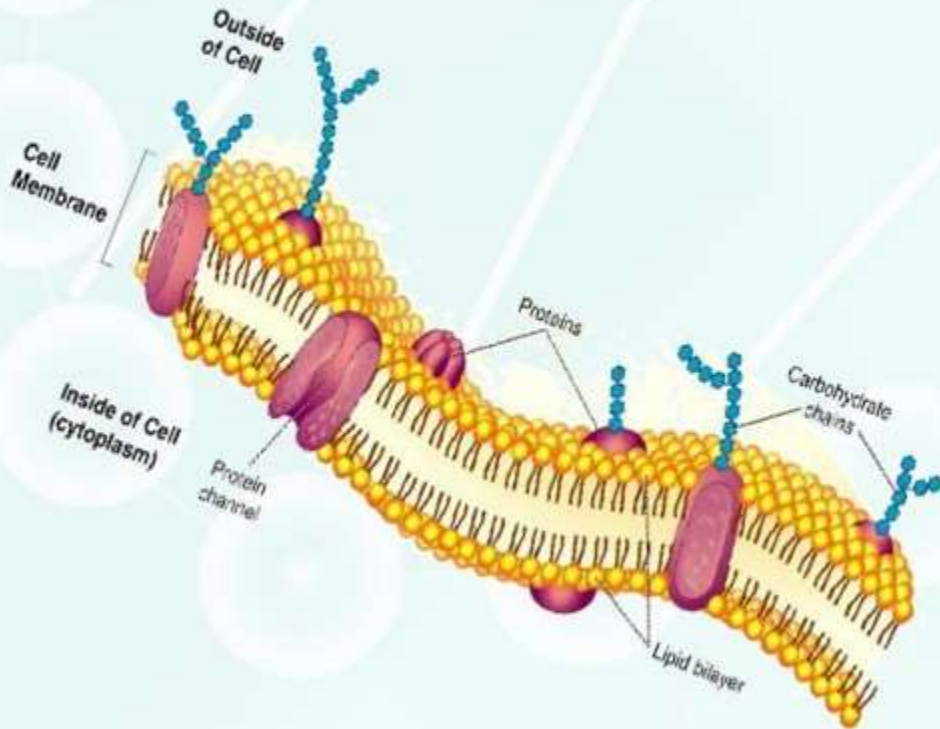


- Bacteria, fungi, algae, plants have a cell wall, but except animals.
- The cell wall of plant cell is made up of cellulose, pectin and lignin etc.
- The bacterial cell wall is made up of peptidoglycan and the fungal wall is made up of chitin.
- Cell wall gives the support and rigidity to the cells.
- In plants the cell walls are structurally and functionally divided into primary cell wall, secondary cell wall and middle lamella.





THE CELL MEMBRANE



- The cell membrane (plasma membrane) is a thin semi-permeable membrane that surrounds the cytoplasm of a cell.
- Its function is to protect the integrity of the interior of the cell by allowing certain substances into the cell, while keeping other substances out.
- It also serves as a base of attachment for the cytoskeleton, thus the cell membrane also serves to help support the cell and help maintain its shape.





CELL ... MOVEMENT OF WATER

RAISINS KEPT IN DIFFERENT CONCENTRATIONS OF WATER



RED BLOOD CELLS KEPT IN DIFFERENT CONCENTRATIONS OF WATER



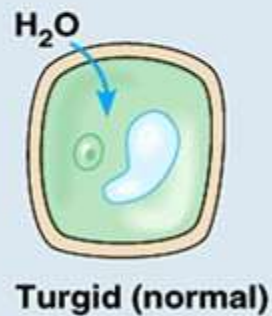
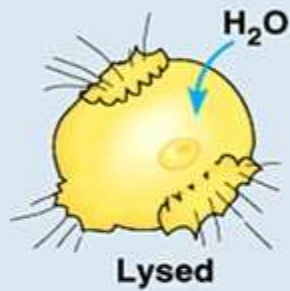
What makes the plant and the animal cell change its shape in different concentration of water ???





OSMOSIS

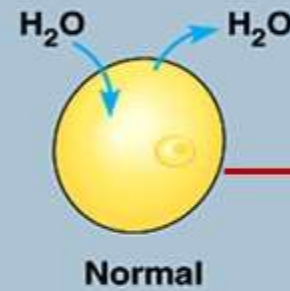
HYPOTONIC SOLUTION



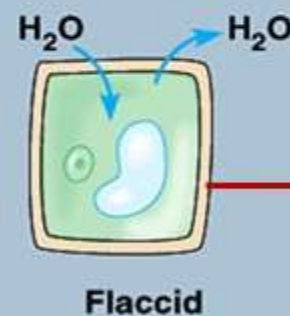
HYPERTONIC SOLUTION



ISOTONIC SOLUTION



→ *Animal cell*



→ *Plant cell.*

The movement of water from its high concentration to its low concentration through a semi-permeable membrane is called osmosis.





THE CYTOPLASM

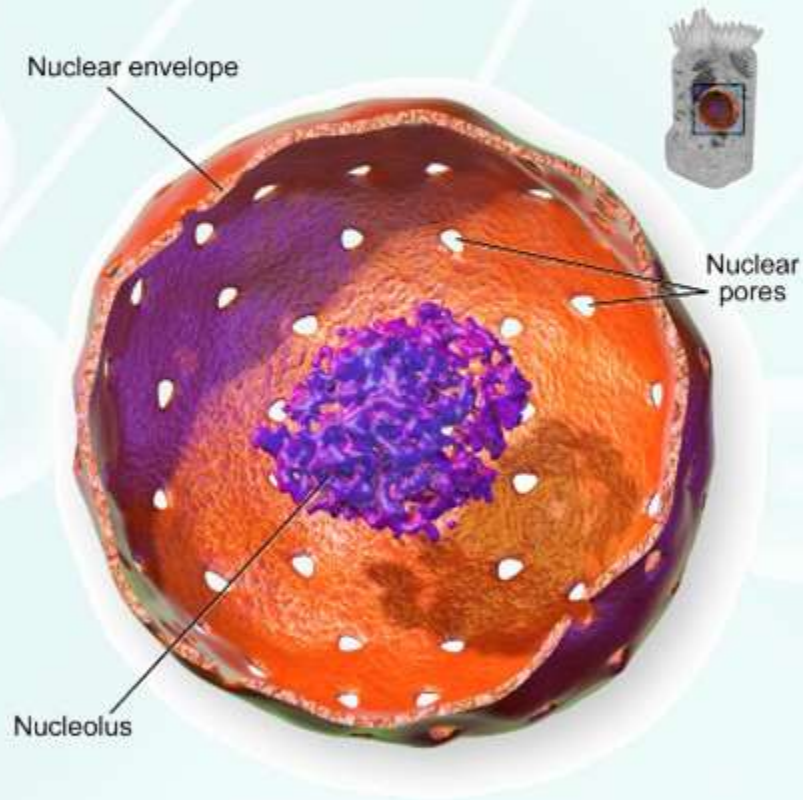


- Cytoplasm is the material inside the cell membrane excluding the nucleus.
- It comprises of gel like cytosol, organelles that are suspended in it and some inclusions.
- Cytosol makes up about 70% of the cell volume and is a complex mixture of cytoskeleton filaments, dissolved molecules, and water.
- The cytosol's filaments include the protein filaments such as actin filaments and microtubules that make up the cytoskeleton, as well as soluble proteins and small structures such as ribosomes, proteasomes etc.
- Small particles of insoluble substances suspended in the cytosol are called inclusions like glycogen, lipids, pigments, crystals etc





THE NUCLEUS



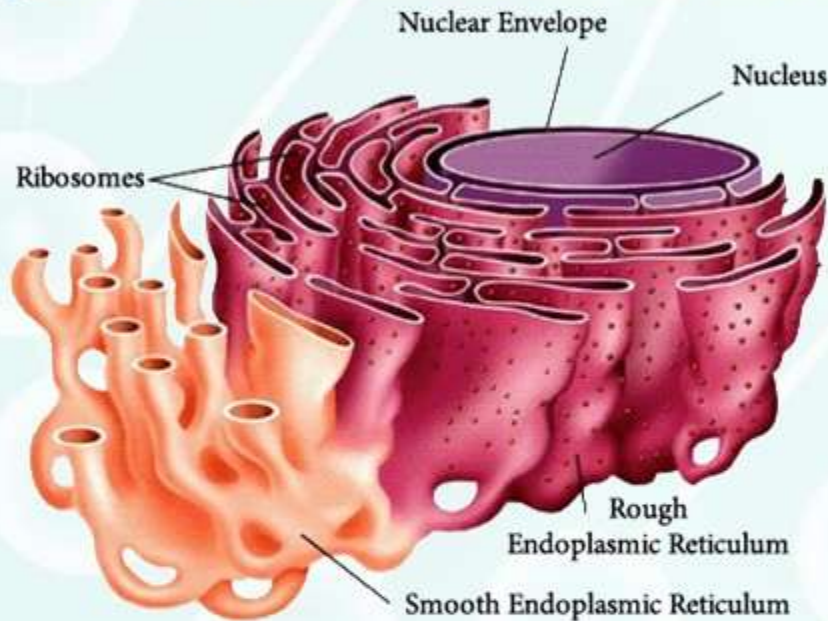
- Also called the Brain of the Cell, the nucleus is the largest cell organelle.
- It is mostly spherical and dense. It has nuclear membrane with nuclear pores for transportation of certain substance
- It is involved in cell division and regulates the cellular functions. If nucleus is removed, the cell dies.
- The nucleus contain thread like structure called chromatin which are made up of DNA.
- One or more round shaped nucleoli inside the nucleus is called the nucleolus.
- Nucleolus participates in protein synthesis by forming and storing RNA.





THE OTHER ORGANELLES- STRUCTURE AND FUNCTION

THE ENDOPLASMIC RETICULUM

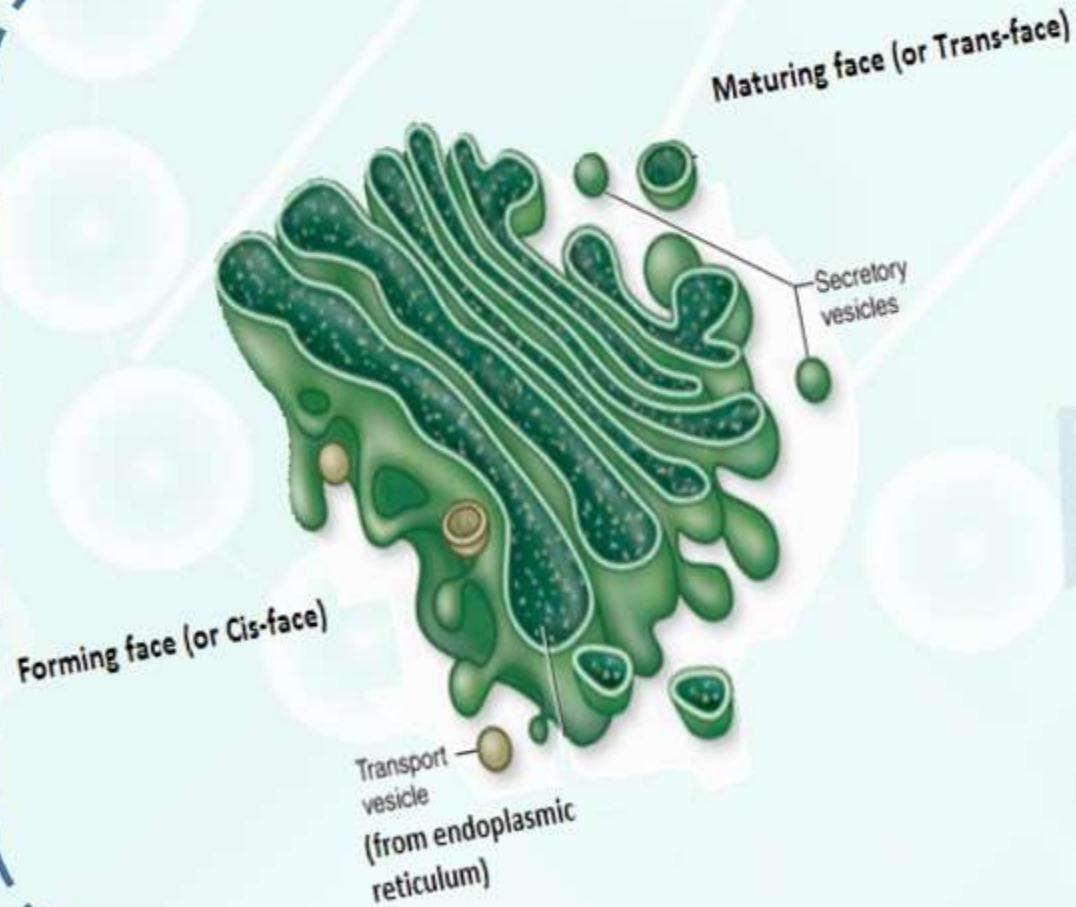


- The ER produces membrane proteins and lipids for its membrane and for many other cell components including lysosomes, secretory vesicles, the Golgi apparatus, the cell membrane, and plant cell vacuoles etc.
- The rough endoplasmic reticulum manufactures membranes and secretory proteins. The ribosomes attached to the rough ER synthesize proteins by the process of translation.
- The smooth endoplasmic reticulum takes part in carbohydrate and lipid synthesis.
- In liver cells the smooth ER produces enzymes that help to detoxify certain compounds.
- In muscles the smooth ER assists in the contraction of muscle cells, and in brain cells it synthesizes male and female hormones





THE GOLGI BODIES/ COMPLEX

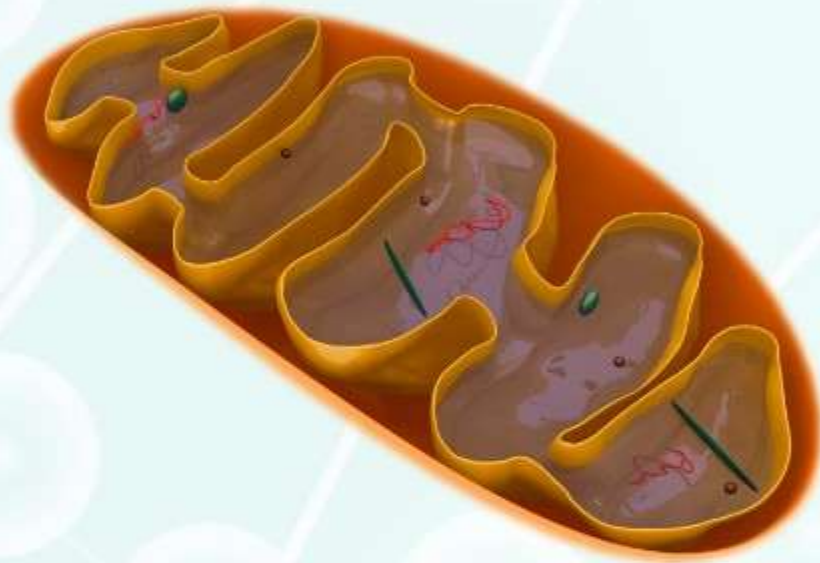


- It was named after the Italian scientist Camillo Golgi who identified it.
- The Golgi apparatus packages proteins into membrane-bound vesicles
- The vesicles are sent to their destination as they detach from the maturing face of the Golgi complex.
- It is of particular importance in processing 'proteins' that is useful for secretion.





THE MITOCHONDRIA

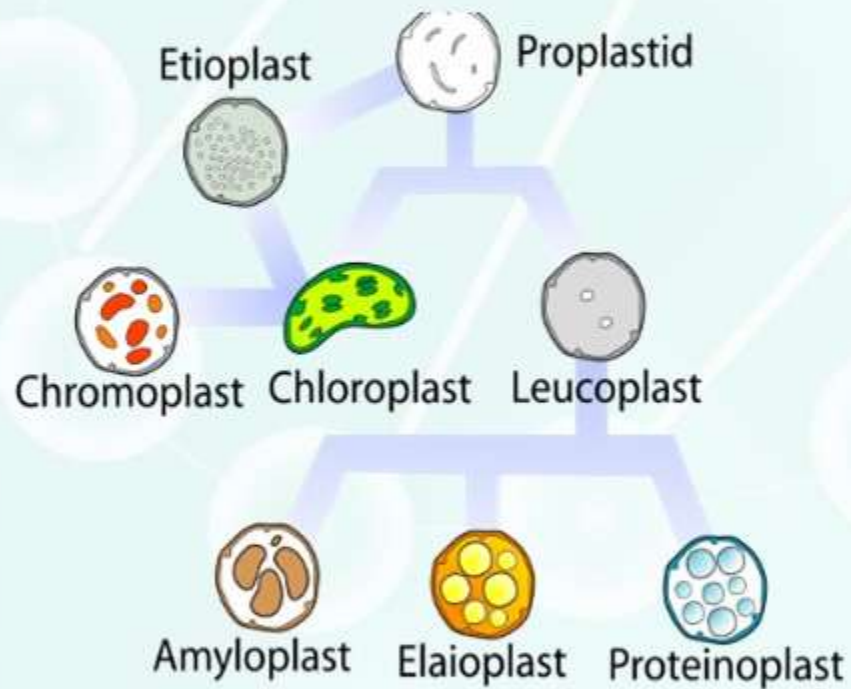


- A mitochondrion is a double membrane organelle found in all eukaryotic organisms(except mature RBC).
- It consists of the outer membrane, the inter-membrane space the inner mitochondrial membrane,
- The infoldings of the inner membrane is the cristae space and the space within the inner membrane is the matrix.
- Mitochondria stripped of their outer membrane are called mitoplasts.
- Mitochondria generate ATP, used as a source of chemical energy in cellular processes.
- Mitochondria has its own independent genome.





THE PLASTIDS

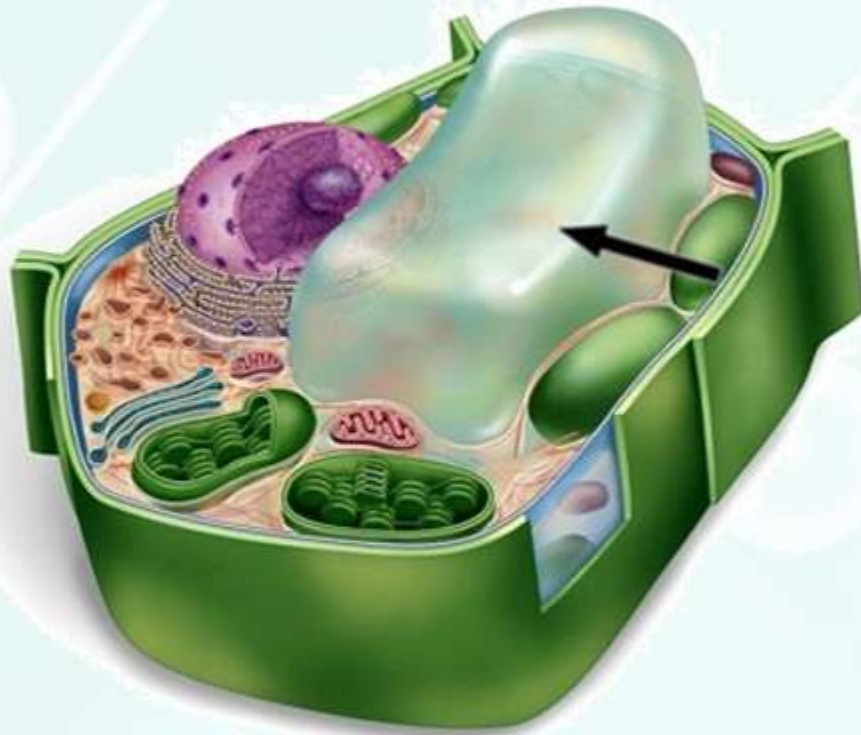


- A major double membrane organelle present in plants, algae etc.
- Plastids are site of manufacture and storage of important chemical compounds in the cell.
- Plastids are autonomous; they possess a double stranded circular DNA.
- In Plants, plastids are differentiated from Proplastid.
- **Chromoplast** - stores the colored pigments
- **Chloroplast** – green pigment for photosynthesis.
- **Leucoplast** – stores different organic compounds like:
 - **Amyloplast** – for starch storage
 - **Elaioplast** – for storage of fats
 - **Proteinoplast** – storage of proteins





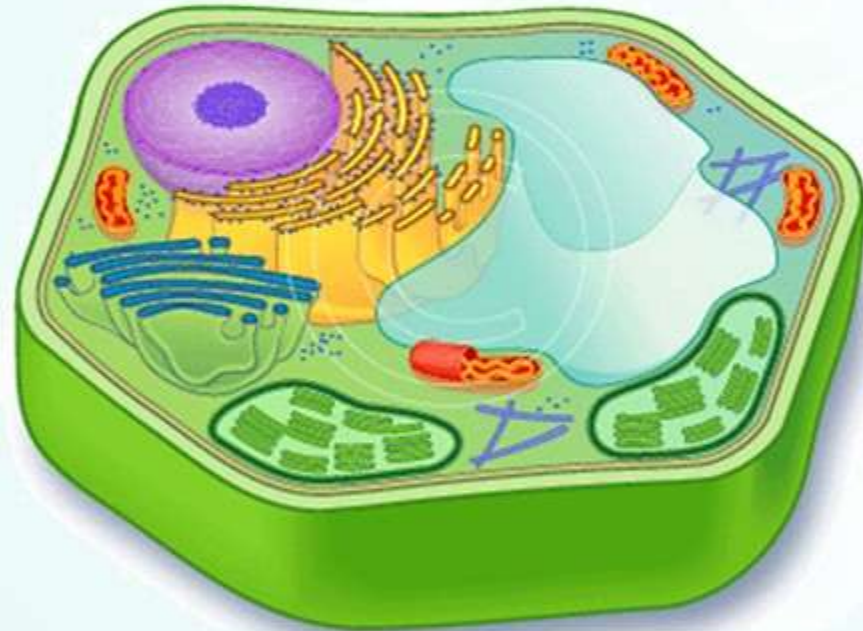
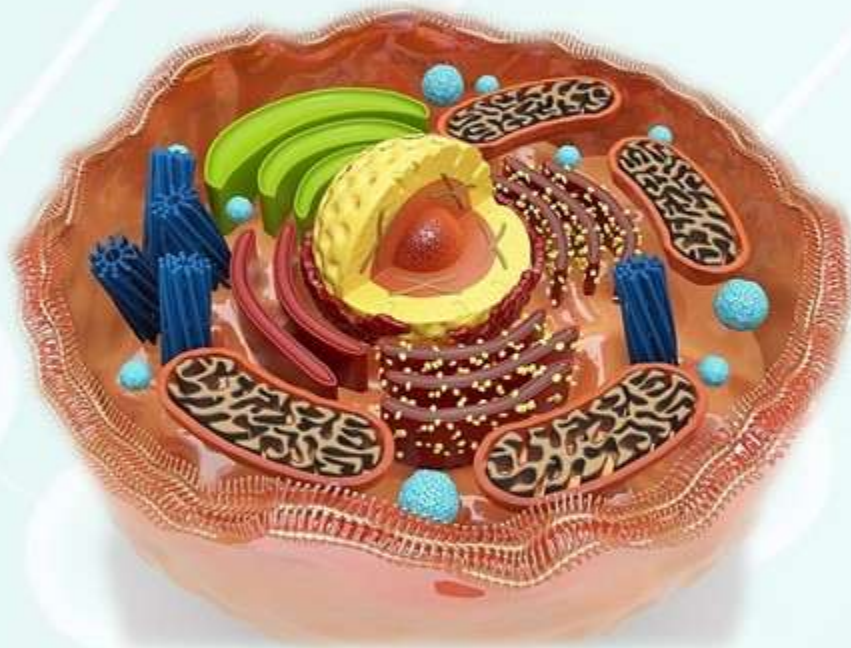
THE VACUOLES



- Vacuoles are clear spaces filled with water or other substances like food, pigments and waste products.
- They are the storage organs of raisins, gum etc.
- Vacuoles are bound by membranes called tonoplast and filled with cell sap.
- Plant cells have larger vacuoles which give them the turgidity (swollen structure).
- Animal cells either have smaller vacuoles or no vacuole at all.



THE COMPARATIVE DIFFERENCE



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